

Geminio: Evaluating VLM Robustness as a Defense Against Gradient Inversion Attacks

Progress Report — April 1, 2026

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1. Summary of Prior Work (v1 Report, March 16)

We reproduced and extended the Geminio attack (Shan et al., 2411.14937) — a VLM-guided gradient inversion attack in federated learning. The attacker (a malicious server) uses CLIP to identify which images match a text query, trains a rigged model that amplifies gradients from those images, then reconstructs them from the victim's gradient updates.

Key findings from the first phase:

- **Cross-domain extension:** Loss ratios of 16x (ImageNet), 8-12x (medical/ChestMNIST), 91x (UAV/drone imagery)
- **Visual distinctiveness drives vulnerability:** visually distinct objects (pools, solar panels) are far easier to target than visually uniform data (chest X-rays)
- **Descriptive prompts improve medical targeting by 39%** (Ahmed's suggestion)
- **Pseudo-labels work:** even 29.6% accuracy labels enable effective attacks
- **Standard defenses fail:** gradient pruning and noise injection are inconsistent
- **53 total ablation experiments** across batch composition, temperature, pseudo-labels, seeds, and batch size

Full details: `reports/detailed_report.md`

2. Direction Shift: Defense Evaluation

2.1 March 16 Meeting

Attendees: Dan Zimmerman, Dr. Ahmed Imteaj, Dr. George Sklivanitis, Md Zarif Hossain.

George and Ahmed agreed that the publishable contribution is **demonstrating effective defenses** against Geminio, not just attack reproduction. George: *"I think it would be of interest to show if you can show that you can reduce significantly the reconstruction capability of the attacker."*

Action items from meeting:

- Pivot from attack exploration to defense evaluation
- Test existing defense mechanisms more thoroughly
- Integrate Ahmed/Zarif's **Federated FAML** framework (CVPR findings track) as a defense
- Collaborate with Zarif on advanced defense mechanisms

2.2 Email Thread Timeline

Date	Event
March 8	Dan emails Ahmed/Zarif with Geminio status update and FathomNet competition parallels
March 9	Ahmed responds: suggests descriptive prompts + UAVScenes dataset. Proposes March 16 meeting
March 9	George confirms availability for March 16
March 16	Meeting: pivot to defense mechanisms, Federated FAML direction
March 24	Zarif shares Sim-CLIP repository (https://github.com/speedlab-git/SimCLIP) and asks Dan to evaluate whether robust VLMs defend against Geminio
April 1	Dan completes SimCLIP/FARE evaluation. Results below.

2.3 SimCLIP/FARE Context

Zarif's Sim-CLIP is an **adversarially robust CLIP variant**. It fine-tunes the vision encoder using a symmetric cosine similarity loss with stop-gradient (Siamese architecture) to make embeddings invariant to adversarial input perturbations (PGD/APGD at $\epsilon = 4/255$). FARE is a comparable baseline using L2 embedding distance loss.

Zarif's request: *"Could you please take a look at the repository and try running the Sim-CLIP models? It would be great if you could evaluate whether these robust models are able to defend against the attack you're studying."*

3. Threat Model Analysis

Before running experiments, we identified a **fundamental mismatch** between the two threat models:

	SimCLIP/FARE Defense	Geminio Attack
What's attacked	CLIP's vision encoder (adversarial pixel perturbations)	FL gradient-sharing protocol
Where CLIP runs	On the victim/client (processing inputs)	On the attacker/server (offline targeting tool)
Attack vector	Perturbed images $\rightarrow$ wrong embeddings	Clean images $\rightarrow$ leaked gradients
Defense mechanism	Robust vision encoder resists perturbations	N/A — images are unperturbed

In Geminio, CLIP processes **real, unperturbed images** on the server side to compute similarity scores. It never sees adversarial perturbations. Robustifying the vision encoder guards against an attack that doesn't occur in this setting.

However, robust vision encoders produce **different embeddings** for the same images, which could incidentally affect the attack. This motivated the experimental evaluation.

4. Experimental Setup

4.1 VLM Variants

Variant	Architecture	Dim	Training	Source
BiomedCLIP	ViT-B/16	512	Domain-specific medical pretraining	microsoft/ BiomedCLIP
Vanilla CLIP	ViT-L/14	768	Standard CLIP (OpenAI)	openai
SimCLIP-4	ViT-L/14	768	Adversarial fine-tuning, cosine loss, $\epsilon = 4/255$	hossainzarif19/ SimCLIP
FARE-4	ViT-L/14	768	Adversarial fine-tuning, L2 loss, $\epsilon = 4/255$	chs20/fare4-clip

SimCLIP and FARE only modify the **vision encoder**. Text encoder is unchanged (verified: text embedding cosine similarity = 1.000 across all CLIP-based variants). Image embeddings differ meaningfully (CLIP vs SimCLIP-4 = 0.923, CLIP vs FARE-4 = 0.914).

4.2 Integration Pipeline

We built a unified VLM adapter (`prototype/vlm_simclip.py`) supporting all four variants via `open_clip`:

```
# Phase 1: Compute image embeddings with each VLM
python prototype/compute_embeddings.py --domain medical --vlm simclip4
python prototype/compute_embeddings.py --domain uav --vlm fare4 --gpu 0

# Phase 2: Train malicious models using each VLM's similarity scores
python prototype/train_medical.py --all --vlm simclip4 --gpu 0
python prototype/train_uav.py --all --vlm fare4 --gpu 0
```

All models trained identically: Adam lr=1e-3, 5 epochs, T=100, classifier-head only.

4.3 Domains

- **Medical:** ChestMNIST (22,433 images, 15 classes), 5 queries, ResNet18
- **UAV:** UAVScenes (4,029 images, 18 classes), 6 queries, ResNet18

5. Results

5.1 Medical Domain Loss Ratios (Top-10%)

Higher loss ratio = stronger attack = worse for defense.

Query	BiomedCLIP	Vanilla CLIP	SimCLIP-4	FARE-4
Pneumonia	10.83x	9.53x	14.40x	11.34x
Cardiomegaly	7.67x	7.14x	12.07x	9.55x
Effusion	10.10x	6.77x	7.26x	15.30x

Normal	4.95x	5.71x	11.87x	10.07x
Lung mass	—	7.47x	11.70x	12.48x
Average	~8.39x	~7.32x	~11.46x	~11.75x
vs Vanilla CLIP	+15%	baseline	+57%	+61%

Both robust VLMs increase medical attack effectiveness by 57-61% over vanilla CLIP.

5.2 UAV Domain Loss Ratios (Top-10%)

Query	Vanilla CLIP	SimCLIP-4	FARE-4
Swimming pool	126.98x	68.05x	50.40x
Solar panels	173.73x	45.05x	67.04x
Trucks on road	22.64x	82.80x	69.09x
River with bridge	153.51x	32.66x	33.40x
Airport runway	64.50x	208.12x	73.48x
Shipping containers	5.34x	44.54x	31.27x
Average	~91.12x	~80.20x	~54.11x
vs Vanilla CLIP	baseline	-12%	-41%

UAV results are mixed. FARE-4 reduces loss ratios by 41% but they remain extremely high (54x average). SimCLIP-4 is close to vanilla CLIP overall but redistributes effectiveness across queries.

5.3 Summary

Domain	Best for Attacker	Worst for Attacker	Defense Verdict
Medical	FARE-4 (11.75x)	Vanilla CLIP (7.32x)	Robust VLMs HELP the attacker
UAV	Vanilla CLIP (91.12x)	FARE-4 (54.11x)	Partial reduction but attack still viable (54x)

6. Analysis

6.1 Why Robust VLMs Strengthen the Medical Attack

Adversarial training forces the vision encoder to learn **semantically meaningful features** rather than texture shortcuts. For medical images — where all chest X-rays look similar to an untrained model — more semantic features mean the VLM can better distinguish pneumonia from cardiomegaly from normal. This is exactly what the attacker needs: better text-image discrimination in the embedding space.

Vanilla CLIP, trained on internet images, struggles with subtle medical distinctions. SimCLIP and FARE, by learning more robust representations, inadvertently provide the attacker with a better targeting tool.

6.2 Why UAV Is Mixed

The UAV domain already has high visual distinctiveness — pools, runways, and solar panels look nothing alike. Vanilla CLIP achieves 91x average loss ratio without any help. The robust encoders don't add meaningful discrimination here, and may lose some texture-level cues that helped vanilla CLIP on certain queries (e.g., swimming pools: 127x $\rightarrow$ 68x).

6.3 Core Conclusion

VLM adversarial robustness is orthogonal to gradient inversion defense. The attack exploits gradient leakage in the FL protocol. The VLM is merely the attacker's targeting tool. Hardening the tool doesn't close the vulnerability — and for domains where vanilla CLIP is already weak (medical), it actually sharpens the attacker's capability.

7. Artifacts Produced

7.1 Code

File	Purpose
prototype/vlm_simclip.py	Unified adapter for CLIP/SimCLIP-4/2/FARE-4 via open_clip
prototype/ compute_embeddings.py	Unified Phase 1 embedding script (--domain, --vlm flags)
prototype/train_medical.py	Updated with --vlm flag (biomedclip, clip, simclip4, fare4)
prototype/train_uav.py	Updated with --vlm flag (clip, simclip4, fare4)
prototype/dataset_medical.py	Configurable embed_path for VLM-specific embeddings

7.2 Computed Embeddings

File	Shape	VLM
data/medical-biomedclip-test.pt	[22433, 512]	BiomedCLIP (prior)
data/medical-clip-test.pt	[22433, 768]	Vanilla CLIP
data/medical-simclip4-test.pt	[22433, 768]	SimCLIP-4
data/medical-fare4-test.pt	[22433, 768]	FARE-4
uavscenes/uav_clip_embeddings_*.pt	[4029, 768]	Vanilla CLIP (prior)
uavscenes/uav_simclip4_embeddings_*.pt	[4029, 768]	SimCLIP-4
uavscenes/uav_fare4_embeddings_*.pt	[4029, 768]	FARE-4

7.3 Trained Models

- 15 new medical models: 5 queries $\times$ 3 VLMs (clip, simclip4, fare4) in malicious_models_medical_v2/
 - 12 new UAV models: 6 queries $\times$ 2 VLMs (simclip4, fare4) in malicious_models_uav/
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8. DP-SGD Defense Evaluation

8.1 Motivation

Since VLM-side defenses failed (Section 5-6), we implemented DP-SGD — a gradient-level defense with formal privacy guarantees (Abadi et al., CCS 2016). DP-SGD applies:

1. **Gradient clipping:** bound the global L2 norm to C (we use $C = 1.0$)
2. **Calibrated Gaussian noise:** $\sigma = C \times \sqrt{2 \ln(1.25/\delta)} / \epsilon$

Implementation: `prototype/defenses.py`, integrated via `--dpsgd-epsilon` flag in both reconstruction scripts.

8.2 Medical Domain (pneumonia_descriptive)

ϵ (Privacy Budget)	LPIPS ↓	PSNR ↑	Attack F1	Privacy Level
None (baseline)	1.019	10.86	0.625	No defense
0.1	0.788	11.35	0.375	Very strong
1.0	0.804	11.31	0.250	Strong
5.0	0.791	11.41	0.125	Moderate
10.0	0.797	11.28	0.250	Weak
50.0	0.836	11.63	0.625	Very weak

8.3 UAV Domain (swimming_pool)

ϵ	LPIPS ↓	PSNR ↑	Attack F1
None	0.679	12.88	—
0.1	0.703	12.49	0.125
1.0	0.693	12.50	0.125
5.0	0.689	12.39	0.000
10.0	0.685	12.33	0.000
50.0	0.852	12.43	0.000

8.4 UAV Validation (solar_panels)

ϵ	LPIPS ↓	PSNR ↑	Attack F1
0.1	0.721	12.36	0.000
1.0	0.720	12.27	0.125
5.0	0.747	12.55	0.000
10.0	0.728	12.46	0.000
50.0	0.764	12.31	0.375

8.5 Analysis

DP-SGD is the first defense that effectively stops Geminio in our experiments.

1. **Gradient clipping is the key mechanism.** Even $\epsilon = 50$ (minimal noise) stops UAV targeting completely for `swimming_pool`. The clipping bounds the gradient norm, directly disrupting Geminio's gradient amplification.
2. **UAV is more sensitive to DP-SGD than medical.** All UAV epsilon values achieve $F1 \leq 0.125$ (`swimming_pool`) or $F1 \leq 0.375$ (`solar_panels`). Medical requires $\epsilon \leq 5$ to reach $F1 = 0.125$. This is the inverse of undefended vulnerability.
3. **Perceptual metrics improve or hold steady.** Medical LPIPS improves from 1.019 $\rightarrow$ 0.788 under DP-SGD. The defense disrupts targeted gradient amplification but overall reconstruction quality is comparable — the attack loses targeting ability.
4. **Contrast with SimCLIP/FARE.** VLM robustness increased medical attack effectiveness by 57-61%. DP-SGD reduces it by 80% ($F1: 0.625 \rightarrow 0.125$). The defense operates at the correct level — gradients, not embeddings.
5. **Results validated across queries.** `Solar_panels` confirms the `swimming_pool` pattern.

8.6 FedAvg Multi-Epoch Defense

The Geminio paper showed FedAvg with multiple local epochs weakens the attack. We tested this systematically.

Medical (pneumonia_descriptive, lr = 1e-3)

Local Epochs LPIPS ↓ PSNR ↑ Attack F1

0 (FedSGD)	1.019	10.86	0.625
1	0.898	11.15	0.500
2	0.813	11.04	0.875
5	0.918	10.87	0.000
10	0.835	9.59	0.500
20	0.819	10.93	0.000

UAV (swimming_pool, lr = 1e-3)

Local Epochs LPIPS ↓ PSNR ↑ Attack F1

0 (FedSGD)	0.679	12.88	—
1	0.660	10.79	0.125
2	0.669	11.04	0.125
5	0.667	10.92	0.375
10	0.668	10.90	0.250
20	0.671	10.95	0.250

FedAvg provides some defense but is **inconsistent** — medical F1 is non-monotonic (0.000 at 5 epochs, 0.500 at 10). DP-SGD is more reliable and provides formal privacy guarantees.

8.7 Complete Defense Comparison

Defense	Level	Medical F1	UAV F1	Reliable?	Formal Guarantee?
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None (FedSGD)	—	0.625	—	—	No
Gradient pruning 99%	Gradient	inconsistent	inconsistent	No	No
Laplacian noise 1e-1	Gradient	inconsistent	inconsistent	No	No
SimCLIP-4	VLM	+ 57% (worse)	-12%	No	No
FARE-4	VLM	+ 61% (worse)	-41%	No	No
FedAvg 5 epochs	FL protocol	0.000	0.375	Partial	No
FedAvg 20 epochs	FL protocol	0.000	0.250	Partial	No
DP-SGD $\epsilon = 5$	Gradient	0.125	0.000	Yes	Yes (ϵ, δ)-DP
